

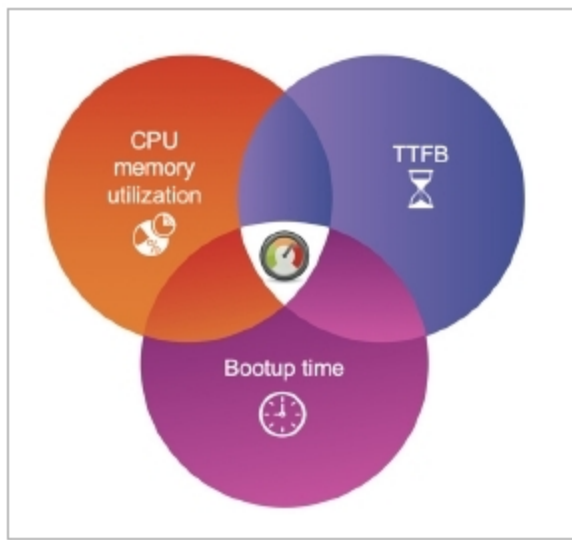


Figure 1: Performance attributes

good to have a real-life example of the services we deploy on a daily basis, rather than a plain 'Hello, world'. Anyway, once I did that, I wanted to compare the numbers for the three parameters, mentioned in Figure 1, where the same service is deployed once using Quarkus native GraalVM, and again in the traditional OpenJDK way using the JVM. When Quarkus bootstraps a project, it also creates a *Dockerfile.native* to deploy the service using GraalVM.

The code of *Dockerfile.native* is shown below:

```
FROM registry.fedoraproject.org/fedora-minimal
WORKDIR /work/
COPY target/*-runner /work/application
RUN chmod 775 /work
EXPOSE 8080
CMD ["/application", "-Dquarkus.http.host=0.0.0.0"]
```

I also created a simple Dockerfile to deploy the service the traditional way, as shown below:

```
FROM openjdk:8-jre-slim
RUN mkdir /app
COPY target/lib /app/lib
COPY target/*-runner.jar /app/application.jar
EXPOSE 8080
CMD ["java", "-jar", "/app/application.jar"]
```

I first ran the traditional deployment and then ran Docker stats, which gave

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIOS
c81f3682b2f1	jovial_chandrasekhar	0.21%	63.98MiB / 1.952GiB	3.20%	891B / 0B	0B / 0B	38

Figure 2: CPU memory utilisation

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIOS
c81f3682b2f1	jovial_chandrasekhar	1.82%	76.54MiB / 1.952GiB	3.83%	12.3kB / 1.95kB	0B / 0B	31

Figure 3: CPU usage

me the data about the CPU and memory utilisation, as shown in Figure 2.

As I hit the service `http://localhost:8080/hello/person`, I saw a peak in CPU usage of 1.82 per cent, which returned to 0.21 per cent after the request was served (Figure 3).

Then, using the Chrome developer tools, I collected the TTFB value as shown in Figure 4.

So the data collected for running the service deployed through the traditional

JVM was as follows.

Memory usage: 64Mib
CPU usage: 0.21 per cent (on ideal state)
CPU usage: 1.82 per cent (on service hit)
TTFB: 557ms

Then I tried deploying the same service in native GraalVM, by running the *Dockerfile.native* created by bootstrapping Quarkus. Figure 5 shows the Docker stats as output.

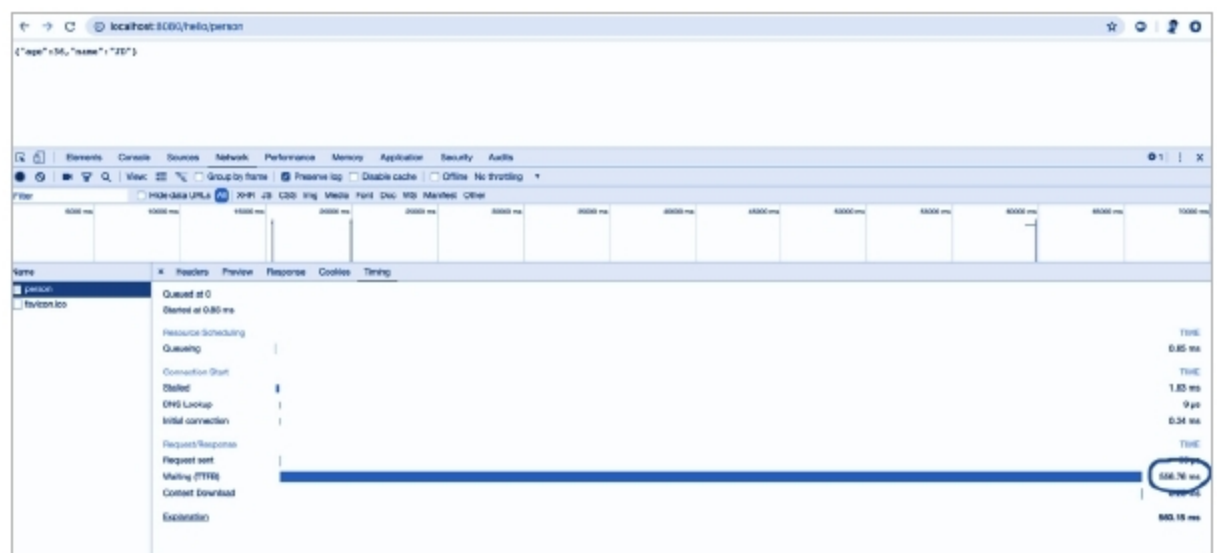


Figure 4: Time to first byte (TTFB)

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIOS
4a1322165186	romantic_feynman	0.00%	2.762MiB / 1.952GiB	0.14%	0B / 0B	0B / 0B	15

Figure 5: CPU memory utilisation (Quarkus)

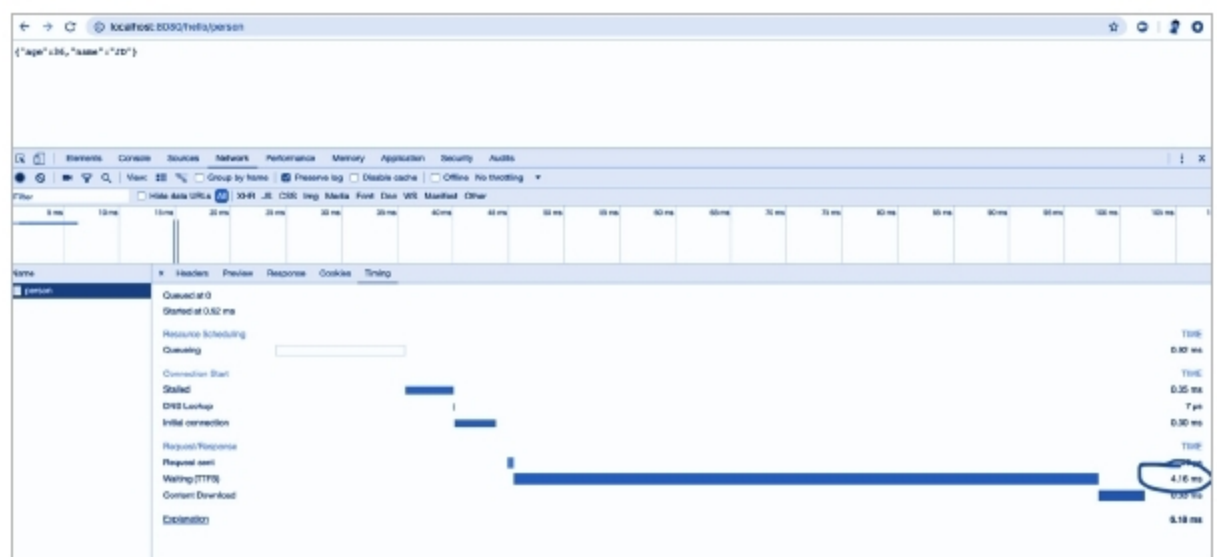


Figure 6: Time to first byte (TTFB) - Quarkus

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